EPFD-Aware Beam Reassociation for Service Recovery in Multi-Beam LEO Satellite Systems under NGSO–GSO Coexistence

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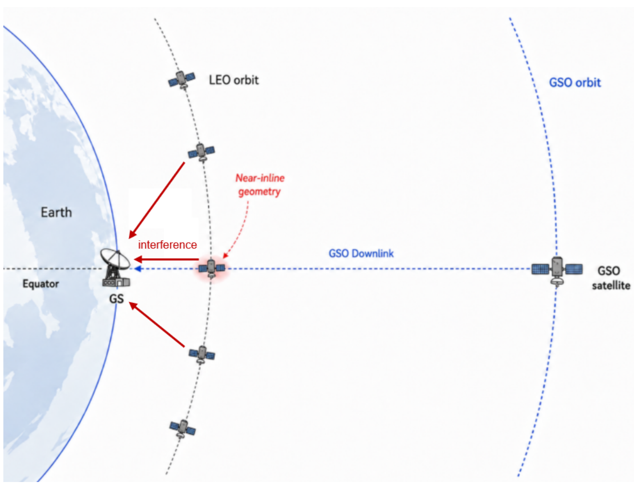


Fig. 1. NGSO–GSO coexistence scenario and near-inline interference geometry between a LEO satellite, a GSO ground station, and the target GSO satellite.

I. Introduction

In recent years, non-geostationary satellite orbit (NGSO) systems, especially low Earth orbit (LEO) constellations, have become an important component of broadband satellite communications, non-terrestrial networks (NTN), and future 6G architectures [1], [2], [7], [8]. Unlike geostationary satellite orbit (GSO) systems, which are located above the Earth's equator and appear fixed with respect to the ground, LEO satellites move rapidly along their orbits and can provide lower latency and more flexible coverage. However, when NGSO downlink transmissions operate in the same or adjacent frequency bands as existing GSO systems, they may cause co-channel interference to GSO ground stations. In particular, when a LEO satellite passes close to the line-of-sight direction between a GSO ground station and its target GSO satellite, a near-inline geometry may occur, leading to a higher interference risk, as illustrated in Fig. 1.

To protect existing GSO systems from interference generated by NGSO systems, the International Telecommunication Union (ITU) specifies interference limits for NGSO systems in Article 22 of the Radio Regulations [9]. Among these limits, equivalent power flux-density (EPFD) is used to evaluate the aggregate interference generated by NGSO transmissions toward GSO systems. Since EPFD depends on satellite geometry, transmit power, antenna gain, propagation loss, and the off-axis angle observed at the GSO ground station, its value changes over time as LEO satellites move along their orbits. Therefore, an NGSO system must ensure that the aggregate EPFD received at the GSO ground station remains below the applicable EPFD limit.

To satisfy the EPFD constraint, NGSO systems may adopt different interference mitigation strategies, such as transmit power control, satellite tilt or pitch adjustment, beam shutdown, and beam steering [17]–[19]. These approaches can reduce the interference observed at the GSO ground station by lowering the transmitted power toward the GSO direction, changing the satellite pointing geometry, or avoiding high-risk beam transmissions. However, such mitigation actions may also reduce the available service resources for LEO users. For example, reducing beam power can decrease the achievable downlink rate, while shutting down high-risk beams can directly interrupt the service of users originally covered by those beams. Therefore, NGSO–GSO coexistence should not only be evaluated from the perspective of EPFD compliance, but also from the perspective of service continuity and user satisfaction.

In a multi-beam LEO system, the service impact of EPFD mitigation becomes more critical during near-inline events. When a LEO satellite passes close to the GSO ground station and the GSO line-of-sight direction, its beams may contribute a large portion of the aggregate EPFD. To satisfy the EPFD limit, the system may need to shut down some high-risk beams on the dominant interfering satellite. Although beam shutdown is effective in reducing the EPFD contribution toward the GSO ground station, it also interrupts the service of users originally served by those closed beams. Therefore, the users under closed beams become the main service demand that must be recovered during the critical period.

This service recovery problem creates an opportunity for beam reassociation. LEO satellite motion and beam footprints are predictable, so the coverage relationship among neighboring satellites can be estimated in advance. In particular, same-orbit neighboring satellites may provide sufficient coverage overlap with the dominant interfering satellite during a critical period. Therefore, users affected by closed-beam shutdown may still be covered by neighboring LEO beams and can potentially be reassociated to these beams for service recovery. This observation motivates the use of neighboring satellites as helper satellites for EPFD-aware service recovery through beam reassociation.

Based on this observation, this work proposes EPFD-Aware Beam Reassociation for Service Recovery (EABR), a beam reassociation framework for multi-beam LEO satellite systems under NGSO–GSO coexistence. The main idea is to exploit the predictable coverage overlap among neighboring LEO satellites. When high-risk beams are shut down to satisfy the EPFD limit, users originally served by those beams may still be covered by nearby helper beams. The proposed EABR framework therefore uses beam reassociation to transfer affected users from closed beams to feasible helper recovery beams, so that their service can be continuously supported during the critical period. In this way, EABR aims to improve user satisfaction while maintaining EPFD compliance.

The main contributions of this work are summarized as follows.

- We propose EPFD-Aware Beam Reassociation for Service Recovery (EABR), a beam reassociation framework for EPFD-constrained multi-beam LEO systems. EABR exploits predictable coverage overlap among neighboring LEO satellites to continuously support users affected by critical-beam shutdown.
- We design the core components of the proposed framework, including a time-slot beam role record, Safe-Beam Reassociation for Helper Users (SBR), released power allocation for helper recovery beams, and Helper-Beam Reassociation for Closed-Beam Users (HBR). The time-slot beam role record captures critical satellites, helper satellites, and overlap-based candidate beam relation graphs. SBR uses critical safe beams to reassociate selected helper users and release helper satellite transmit power. The released power is then allocated to helper recovery beams, while HBR reassociates closed-beam users to feasible helper recovery beams.
- We evaluate the proposed framework using an STK–MATLAB simulation framework [25], [26]. The results show that the proposed framework satisfies the EPFD constraint while improving average user satisfaction, enhancing service recovery capability, and reducing the proportion of low-satisfaction users compared with baseline methods.

II. Related Work

Existing studies on NGSO–GSO coexistence have investigated several interference mitigation strategies for reducing NGSO downlink interference toward GSO systems. In this chapter, related works are reviewed in three categories: progressive pitch and beam shutdown, joint power and tilt control, and beamforming-based interference mitigation. The characteristics and limitations of these methods are discussed to clarify the motivation of this work.

A. Progressive Pitch and Beam Shutdown

Progressive pitch is one representative attitude-based interference avoidance strategy for NGSO–GSO coexistence. In this approach, the LEO satellite gradually adjusts its attitude when it approaches low-latitude regions, where its downlink direction may become close to the line-of-sight direction from the GSO ground station to the target GSO satellite. By changing the satellite pointing direction, the LEO downlink beam can be shifted away from this line-of-sight direction, thereby increasing the angular separation observed at the GSO ground station. When attitude adjustment alone is insufficient to satisfy the interference constraint, some high-risk user beams may be shut down to further reduce interference.

Li et al. [19] studied the progressive pitch strategy of the OneWeb constellation. Their results show that progressive pitch can effectively reduce interference to GSO systems in many regions. However, the combination of satellite tilting and partial beam shutdown may also reduce coverage continuity and create service blind areas during certain periods. This indicates that interference avoidance based only on attitude adjustment and beam shutdown may still affect the service availability of LEO users.

B. Joint Power and Tilt Control

Another type of interference mitigation method combines transmit power control with satellite tilt control. Power control reduces the transmit power of LEO satellites when they approach high-risk regions near the GSO ground station, so that the interference received at the GSO ground station can be lowered. Compared with progressive pitch methods that follow a predefined attitude adjustment strategy, joint power and tilt control formulates the mitigation process as an optimization problem. The transmit power and tilt angle of LEO satellites are jointly adjusted to satisfy the EPFD constraint while improving user demand satisfaction.

Jalali et al. [18] proposed a joint power and tilt control method for NGSO–GSO interference mitigation. Their method optimizes LEO transmit power and satellite tilt angle under the aggregate EPFD constraint. Compared with direct beam shutdown, this approach provides a more flexible way to reduce interference and preserve user service. However, when the interference constraint is strict, the transmit power of critical satellites may still need to be significantly reduced, which can limit the achievable service quality of LEO users during critical periods.

C. Beamforming and Beam Steering

Beamforming-based methods provide another interference mitigation direction for NGSO satellite systems equipped with phased-array antennas. Instead of only reducing transmit power or adjusting satellite attitude, beamforming controls antenna weight vectors to steer signals toward intended LEO users and suppress emissions toward GSO ground stations. In this way, the NGSO system can improve its downlink transmission while reducing co-channel interference to the GSO system.

Jalali et al. [20] investigated downlink beamforming strategies for interference-aware NGSO satellite systems. They proposed an aggregate interference-constrained beamforming method to limit the interference received at GSO ground stations while satisfying the service requirements of LEO users. This approach shows that beamforming can be an effective interference mitigation technique for advanced NGSO payloads. However, beamforming generally requires phased-array antennas, channel state information or average channel state information, and higher computational complexity.

Since this work focuses on a fixed multi-beam LEO system, beamforming-based control is not considered in this thesis.

D. Summary and Research Gap

The above studies show that NGSO–GSO interference can be mitigated through satellite attitude adjustment, transmit power control, beam shutdown, or antenna-domain beamforming. These methods reduce the interference received at GSO ground stations from different perspectives. However, when mitigation actions reduce the available resources of LEO satellites, the service quality of LEO users may still be affected. In particular, users originally served by beams that are shut down during critical periods require additional service recovery support.

Existing studies have not addressed how neighboring LEO satellite beams can be used to recover the service of users affected by critical-beam shutdown under the GS EPFD constraint. This motivates the proposed EABR framework, which exploits predictable coverage overlap among neighboring LEO satellites for service recovery while maintaining EPFD compliance.

III. System Model and Problem Formulation

A. System Model

1) Network Scenario: We consider a general multi-beam LEO satellite communication system under NGSO–GSO coexistence. Since the LEO downlink and the GSO system may operate in the same frequency band, downlink transmissions from visible LEO satellites can generate co-channel interference at a GSO ground station. Hereafter, this ground station is referred to as the GS. The GS communicates with its target GSO satellite, and the direction from the GS to the target GSO satellite is used as the reference direction for EPFD evaluation. Among the LEO satellites visible from the GS, a near-inline NGSO–GSO geometry may occur when a satellite has a downlink beam whose footprint covers the GS and simultaneously passes close to the GS-to-GSO reference direction as observed from the GS, resulting in a higher interference risk.

The system is modeled in discrete time slots. Let $\mathcal{T}$ denote the set of time slots considered in this work, and let $t \in \mathcal{T}$ denote a specific time slot. For each time slot t, the satellite geometry, beam footprints, user coverage, and EPFD contribution may change due to LEO satellite motion. Therefore, the system state is evaluated on a time-slot basis.

Let $\mathcal{S}(t)$ denote the set of LEO satellites visible from the GS at time slot t, and let $\mathcal{B}_s$ denote the set of beams on satellite s. Each satellite employs a satellite-fixed multi-beam downlink layout, where each beam has a predefined pointing direction relative to the satellite and forms a corresponding ground footprint. For beam b on satellite s, the ground footprint at time slot t is denoted by $\mathcal{F}_{s,b}(t)$. If user u lies within $\mathcal{F}_{s,b}(t)$, beam b on satellite s is regarded as a feasible serving candidate for that user.

Because neighboring LEO satellites and their beams may have overlapping ground footprints, a terrestrial user may be covered by multiple beams from visible LEO satellites in the same time slot. This overlap provides multiple feasible serving beams for user association and resource coordination.

In this work, these feasible beams are assumed to operate with compatible downlink frequency and polarization settings. Therefore, changing the serving beam of a user can be modeled as a serving-beam reassociation among the beams that cover the user. This assumption is reasonable because predictable satellite motion, overlapping beam coverage, and compatible radio resources can support service continuity when a user is reassociated to another feasible beam [3]–[6].

2) User Association and Service Model: Let $\mathcal{U}(t)$ denote the set of terrestrial users considered at time slot t. Each user can be served by at most one beam in the same time slot. The user association is represented by a binary indicator $x_{u,s,b}(t)$, defined as

$$x_{u,s,b}(t) = \begin{cases} 1, & \text{if user } u \text{ is served by beam } b \text{ of satellite } s, \\ 0, & \text{otherwise,} \end{cases}$$

for all $u \in \mathcal{U}(t)$, $s \in \mathcal{S}(t)$, and $b \in \mathcal{B}_s$. A user can only be associated with a beam that covers its location. Therefore, if user u is not located within the footprint $\mathcal{F}_{s,b}(t)$, beam b of satellite s is not a feasible serving beam for this user.

For beam b of satellite s, let $N_{s,b}(t)$ denote the number of users served by this beam at time slot t. This value represents the beam loading, which reflects how many users share the available downlink resource of the same beam.

Each user u has a service demand. The demand represents the required downlink data rate of the user. In addition, different users may have different service priorities. Motivated by the priority concept in 3GPP QoS differentiation [15], each user u is assigned a priority class $\pi(u)$, and each priority class is associated with a priority weight $w_{\pi(u)}$. A higher priority weight indicates that the user has a higher importance in service recovery and resource allocation.

The user association state at time slot t is denoted by $\mathbf{X}(t)$, which contains all association indicators $x_{u,s,b}(t)$. The association decisions across all time slots are denoted by $\mathbf{X} = \{\mathbf{X}(t) \mid t \in \mathcal{T}\}$.

3) Beam Power Model: Each beam is assigned a transmit power in each time slot. Let $p_{s,b}(t)$ denote the transmit power of beam b on satellite s at time slot t. The beam power allocation state at time slot t is denoted by $\mathbf{P}(t)$, which contains all beam transmit powers $p_{s,b}(t)$. The beam power allocation decisions across all time slots are denoted by $\mathbf{P} = \{\mathbf{P}(t) \mid t \in \mathcal{T}\}$.

Under normal operation, each beam is assigned a predefined default transmit power. The default beam-power profile is configured according to beam geometry, antenna gain, and propagation loss. The purpose is to provide ground users in different beam footprints with similar received downlink power. This default setting represents the baseline power allocation before EPFD-aware power adjustment or beam shutdown is applied.

Each beam has a beam-level power upper bound, denoted by $P_{s,b}^{\max}$, and each satellite has a total transmit power budget, denoted by $P_{s,tot}^{\max}$. During EPFD-aware operation, the transmit power of a beam may be adjusted from its default value. A beam is regarded as shut down when its transmit power is set to zero. The beam-level and satellite-level power limits are included later in the problem formulation as power constraints.

4) SINR Model: The downlink SINR of a user is modeled using a standard wireless and satellite communication formulation, where the received desired signal, LEO co-channel interference, and receiver noise are considered. If user u is served by beam b of satellite s at time slot t, its downlink SINR is given by

$$\mathrm{SINR}_u(t) = \frac{S_u(t)}{I_u^{\mathrm{intra}}(t) + I_u^{\mathrm{inter}}(t) + N_u(t)}, \tag{1}$$

where $S_u(t)$ is the desired signal power, $I_u^{\mathrm{intra}}(t)$ is the LEO intra-satellite co-channel interference, $I_u^{\mathrm{inter}}(t)$ is the LEO inter-satellite co-channel interference, and $N_u(t)$ is the receiver noise power.

Desired Signal Power: The desired signal power follows the standard satellite link-budget relation. For user u served by beam b of satellite s, the received desired signal power is given by

$$S_u(t) = \frac{p_{s,b}(t)\, G^{\mathrm{tx}}_{s,b\to u}(t)\, G^{\mathrm{rx}}_u}{L_{s,u}(t)},$$

where $p_{s,b}(t)$ is the transmit power of beam b on satellite s, $G^{\mathrm{tx}}_{s,b\to u}(t)$ is the transmit antenna gain of beam b toward user u, G^{rx}_u is the user terminal receive gain, and $L_{s,u}(t)$ is the total propagation loss between satellite s and user u.

The total propagation loss includes free-space path loss and atmospheric gaseous loss:

$$L_{s,u}(t) = L^{\mathrm{fs}}_{s,u}(t)\, L^{\mathrm{gas}}_{s,u}(t),$$

where $L^{\mathrm{fs}}_{s,u}(t)$ is the free-space path loss and $L^{\mathrm{gas}}_{s,u}(t)$ is the atmospheric gaseous loss. The free-space path loss is given by

$$L^{\mathrm{fs}}_{s,u}(t) = \left(\frac{4\pi d_{s,u}(t)}{\lambda}\right)^2,$$

where $d_{s,u}(t)$ is the distance between satellite s and user u, and λ is the carrier wavelength.

LEO Intra-Satellite Interference: The LEO intra-satellite interference comes from co-channel beams on the same satellite. Let $\mathcal{B}^{\mathrm{co}}_s(b)$ denote the set of beams on satellite s that use the same downlink frequency resource as the serving beam b, excluding beam b itself. The intra-satellite interference received by user u is given by

$$I_u^{\mathrm{intra}}(t) = \sum_{b' \in \mathcal{B}^{\mathrm{co}}_s(b)} \frac{p_{s,b'}(t)\, G^{\mathrm{tx}}_{s,b'\to u}(t)\, G^{\mathrm{rx}}_u}{L_{s,u}(t)}.$$

LEO Inter-Satellite Interference: The LEO inter-satellite interference comes from co-channel beams on other visible LEO satellites. For another visible satellite s', let $\mathcal{B}^{\mathrm{co}}_{s'}(b)$ denote the set of beams on satellite s' that use the same downlink frequency resource as the serving beam b. The inter-satellite interference received by user u is given by

$$I_u^{\mathrm{inter}}(t) = \sum_{\substack{s' \in \mathcal{S}(t) \\ s' \neq s}} \sum_{b' \in \mathcal{B}^{\mathrm{co}}_{s'}(b)} \frac{p_{s',b'}(t)\, G^{\mathrm{tx}}_{s',b'\to u}(t)\, G^{\mathrm{rx}}_u}{L_{s',u}(t)},$$

where $G^{\mathrm{tx}}_{s',b'\to u}(t)$ is the transmit antenna gain of the interfering beam toward user u, and $L_{s',u}(t)$ is the total propagation loss between satellite s' and user u.

With this SINR model, the user SINR changes when the serving beam, beam transmit power, propagation loss, or LEO co-channel interference changes.

5) Capacity and User Satisfaction Model: After obtaining the user SINR, the achievable downlink capacity is modeled by the Shannon capacity formula, which is commonly used in wireless and satellite communication systems. Let B denote the downlink bandwidth of a beam. If user u is served by beam b of satellite s at time slot t, the achievable downlink capacity of user u is given by

$$C_u(t) = B \log_2\!\big(1 + \mathrm{SINR}_u(t)\big).$$

When multiple users are served by the same beam, they share the downlink resource of that beam. In this work, users served by the same beam are assumed to equally share the beam resource in each time slot. The actual service rate of user u is therefore

$$R_u(t) = \frac{C_u(t)}{N_{s,b}(t)}, \tag{2}$$

where $N_{s,b}(t)$ represents the beam loading of beam b on satellite s at time slot t. This means that users served by the same beam are allocated the same fraction of the available beam resource.

Let D_u denote the service demand of user u. Following the demand satisfaction concept used in [18], the satisfaction of user u is calculated by comparing the actual service rate with the user demand and is defined as

$$s_u(t) = \min\left(\frac{R_u(t)}{D_u}, 1\right), \tag{3}$$

where $s_u(t) \in [0, 1]$. If $s_u(t) = 1$, the demand of user u is fully met; if $0 < s_u(t) < 1$, user u receives partial service; if $s_u(t) = 0$, user u receives no effective service in the time slot. This satisfaction metric represents the fraction of user demand that is satisfied in each time slot.

6) EPFD Model: The EPFD calculation follows the ITU Radio Regulations and ITU-R S.1503 framework for NGSO–GSO coexistence [9], [10]. In this work, the EPFD at the GS is calculated by first computing the contribution of each active beam from visible LEO satellites. For each active beam, the contribution depends on its transmit power, transmit antenna gain toward the GS, propagation distance, and the receive antenna gain of the GS. The contributions of all active beams from all visible LEO satellites are then summed in the linear domain to obtain the aggregate EPFD at the GS. This calculation considers each active beam separately because different beams on the same LEO satellite may have different pointing directions, antenna gains, and transmit powers toward the GS.

For EPFD calculation, let $P^{\mathrm{ref}}_{s,b}(t)$ denote the radio-frequency (RF) transmit power of beam b on satellite s within the ITU reference bandwidth B_{ref} specified for the applicable EPFD limit [11], [12]. If the beam transmit power $p_{s,b}(t)$ is defined over a wider beam bandwidth, it is converted to the corresponding RF power within $B_{\rm ref}$ before applying the EPFD formula. The EPFD contribution of beam b on satellite s toward the GS is given by

$$\mathrm{EPFD}_{s,b}(t) = \frac{P^{\rm ref}_{s,b}(t)\, G^{\rm tx}_{s,b\to g}(t)}{4\pi d^2_{s,g}(t)} \cdot \frac{G^{\rm rx}_g(\theta_{g\to s}(t))}{G^{\rm rx}_{g,\max}},$$

where $G^{\rm tx}_{s,b\to g}(t)$ is the transmit antenna gain of beam b on satellite s in the direction of the GS, $d_{s,g}(t)$ is the distance between satellite s and the GS, $G^{\rm rx}_g(\theta_{g\to s}(t))$ is the receive antenna gain of the reference GS antenna in the direction of satellite s, $G^{\rm rx}_{g,\max}$ is the maximum receive antenna gain of the reference GS antenna, and $\theta_{g\to s}(t)$ is the off-axis angle between the GS antenna boresight and the direction of satellite s.

The transmit antenna gain $G^{\rm tx}_{s,b\to g}(t)$ is evaluated from the beam antenna pattern according to the off-axis angle between the boresight of beam b and the direction of the GS. Since each beam has its own pointing direction, the EPFD contribution is computed separately for each beam. If a beam is shut down, its transmit power is zero and its EPFD contribution is also zero.

The aggregate EPFD at the GS is obtained by summing the EPFD contributions of all active beams of all visible LEO satellites:

$$\mathrm{EPFD}_{\rm agg}(t) = \sum_{s\in\mathcal{S}(t)} \sum_{b\in\mathcal{B}_s} \mathrm{EPFD}_{s,b}(t).$$

The resulting aggregate EPFD is compared with the applicable EPFD limit in the problem formulation.

B. Problem Formulation

Based on the system model above, we formulate the user association and beam power allocation problem under user association, beam power, satellite power budget, and EPFD constraints. The decision variables are the user association sequence $\mathbf{X}$ and the beam power allocation sequence $\mathbf{P}$. Specifically, $\mathbf{X}(t) = \{x_{u,s,b}(t)\}$ denotes the user association state at time slot t, and $\mathbf{P}(t) = \{p_{s,b}(t)\}$ denotes the beam power allocation state at time slot t.

Objective function:

$$\max_{\mathbf{X},\mathbf{P}} \sum_{t\in\mathcal{T}} \sum_{u\in\mathcal{U}(t)} w_{\pi(u)}\, s_u(t). \tag{4}$$

Constraints:

$$\sum_{s\in\mathcal{S}(t)} \sum_{b\in\mathcal{B}_s} x_{u,s,b}(t) \le 1. \tag{5}$$

$$x_{u,s,b}(t) \in \{0,1\}. \tag{6}$$

$$0 \le p_{s,b}(t) \le P^{\max}_{s,b}. \tag{7}$$

$$\sum_{b\in\mathcal{B}_s} p_{s,b}(t) \le P^{\max}_{s,tot}. \tag{8}$$

$$\mathrm{EPFD}_{\rm agg}(t) \le \mathrm{EPFD}_{\rm lim}. \tag{9}$$

where (5) holds for all $u \in \mathcal{U}(t)$ and $t \in \mathcal{T}$; (6) holds for all $u \in \mathcal{U}(t)$, $s \in \mathcal{S}(t)$, $b \in \mathcal{B}_s$, and $t \in \mathcal{T}$; (7) holds for all $s \in \mathcal{S}(t)$, $b \in \mathcal{B}_s$, and $t \in \mathcal{T}$; (8) holds for all $s \in \mathcal{S}(t)$ and $t \in \mathcal{T}$; and (9) holds for all $t \in \mathcal{T}$.

Constraint (5) ensures that each user is served by at most one beam in each time slot. As described in the user association model, only beams that cover the user are considered feasible serving beams.

Constraint (6) defines the binary user association variable.

Constraint (7) limits the transmit power of each beam and includes the possibility of beam shutdown when $p_{s,b}(t) = 0$.

Constraint (8) enforces the total transmit power budget of each satellite.

Constraint (9) ensures that the aggregate EPFD at the GS does not exceed the applicable EPFD limit.

The objective in (4) maximizes the weighted satisfaction of users. The weight $w_{\pi(u)}$ reflects the service priority of user u, while $s_u(t)$ represents the fraction of the user demand that is satisfied in time slot t. Therefore, higher-priority users can receive greater importance in the objective function.

The association and power allocation decisions are strongly coupled. User association affects beam loading, downlink rate, and user satisfaction, while beam power affects both downlink SINR and the aggregate EPFD at the GS. In near-inline NGSO–GSO geometries, shutting down beams can lower the aggregate EPFD, but it may also reduce the service rates of users originally served by the beams that are shut down. Therefore, the problem becomes a mixed-integer nonlinear optimization problem. Solving the global optimum over all time slots is computationally difficult, so the next chapter develops EPFD-Aware Beam Reassociation for Service Recovery to obtain feasible association and power allocation decisions under the EPFD constraint.

IV. Method

A. Method Overview

Chapter III formulates the service recovery problem with user association, beam power, satellite power budget, and EPFD constraints. Under the EPFD constraint, beams from one or more LEO satellites may need to be shut down. A satellite is regarded as critical when at least one of its beams covers the GS and must be shut down to satisfy the EPFD constraint. It should be noted that a critical satellite is not a fixed satellite identity. Instead, this role is assigned according to the satellite geometry and EPFD condition in each time slot. A satellite is not predefined as critical before entering the near-inline period. It is regarded as critical only during the time slots in which its beams must be shut down due to the GS geometry and the EPFD constraint. Once the satellite moves away and its beams no longer need to be shut down for EPFD compliance, the satellite is no longer treated as a critical satellite. Since this criterion is evaluated for all visible LEO satellites, multiple critical satellites may exist in the same time slot. To recover the service of users affected by beam shutdown on critical satellites, this chapter presents an EPFD-aware beam reassociation framework for service recovery.

The proposed framework is designed based on predictable satellite motion, beam footprints, EPFD contributions, and user service requirements. For each time slot, the framework determines which beams need to be shut down to satisfy the EPFD constraint and which neighboring satellites can support service recovery according to the current satellite geometry and EPFD condition. It then constructs candidate beam relation graphs to describe the possible reassociation relations between beams of critical satellites and neighboring helper satellites.

The proposed framework consists of three major online procedures. First, Safe-Beam Reassociation for Helper Users (SBR) uses the residual capacity of critical safe beams to reassociate part of the helper users originally served by helper release beams. This reduces the loading of helper release beams and releases part of the helper satellite transmit power. Second, the released power is allocated to helper recovery beams according to their priority-weighted recovery demand. Third, Helper-Beam Reassociation for Closed-Beam Users (HBR) uses helper recovery beams to recover closed-beam users. Through these procedures, the proposed framework aims to improve service continuity for closed-beam users while maintaining EPFD compliance and power constraints.

In summary, the proposed EPFD-aware beam reassociation framework consists of the following steps:

- **Time-slot beam role record:** Precompute the critical satellites, closed beams, critical safe beams, helper release beam candidates, helper recovery beam candidates, and candidate beam relation graphs for each time slot.
- **Safe-Beam Reassociation for Helper Users (SBR):** Reassociate selected helper users from helper release beams to critical safe beams, reducing helper beam loading and releasing helper satellite transmit power.
- **Released power allocation for helper recovery beams:** Allocate the released transmit power to helper recovery beams according to the priority-weighted recovery demand.
- **Helper-Beam Reassociation for Closed-Beam Users (HBR):** Reassociate selected closed-beam users to helper recovery beams and recover their service under the available power of helper recovery beams.

B. Time-Slot Beam Role Record

Before the online reassociation procedures are executed, the system constructs a time-slot beam role record for each time slot. This record stores the critical satellites, neighboring helper satellites, and the beam pairs that may be used for reassociation in the current time slot. These records are mainly built according to predictable satellite geometry, beam footprints, and EPFD contributions.

1) Critical Satellite and Beam Role Identification: For each time slot t, the system first checks whether the aggregate EPFD generated by all active beams of visible LEO satellites satisfies the EPFD limit. If the aggregate EPFD is already below the limit, beam shutdown is not required in this time slot, and the critical satellite set is empty. If the aggregate EPFD exceeds the limit, the system examines the beam-level EPFD contribution of each active beam on all visible LEO satellites and sorts the beams in descending order according to $\mathrm{EPFD}_{s,b}(t)$. The system then sequentially shuts down beams with higher EPFD contributions and updates the aggregate EPFD after each beam shutdown until the aggregate EPFD satisfies the EPFD limit.

Let $\mathcal{B}^{\mathrm{off}}_{s,l}(t)$ denote the set of beams shut down on satellite s in time slot t. After the beam shutdown process is completed, all satellites with a non-empty $\mathcal{B}^{\mathrm{off}}_{s,l}(t)$ form the critical satellite set, denoted by $\mathcal{S}^{\mathrm{C}}(t)$.

For each critical satellite s, the beams that are not shut down and remain active are denoted by $\mathcal{B}^{\mathrm{safe}}_{s,l}(t)$, and are referred to as safe beams. Therefore, the critical satellite set is determined by the beam shutdown result of each time slot. If beams from different satellites must be shut down to satisfy the EPFD constraint, these satellites are included in $\mathcal{S}^{\mathrm{C}}(t)$ at the same time.

2) Helper Satellite and Beam Role Identification: After critical set identification is completed, the system constructs the helper satellite set, denoted by $\mathcal{S}^{\mathrm{H}}(t)$. Similar to the critical satellite role, the helper satellite is also a time-dependent operational role rather than a permanent satellite identity. A neighboring satellite is regarded as a helper only when its beams have feasible overlap relations with the current critical satellites and can support the service recovery process in the current time slot. Once the corresponding critical satellites no longer need to shut down beams for EPFD compliance, the helper relation is also released. For each helper satellite candidate, the system checks which of its beams overlap with the beams of the current critical satellites. Based on these overlap relations, each helper beam is then classified as a possible helper release beam or helper recovery beam.

If the service area of a helper beam overlaps with the footprint of critical safe beams, this helper beam is marked as a helper release beam candidate, denoted by $\mathcal{B}^{\mathrm{rls}}_{h,k}(t)$. A helper release beam candidate represents a helper beam whose served users overlap with critical safe beams and whose loading may be reduced through reassociation.

If the service area of a helper beam overlaps with the footprint of critical closed beams, this helper beam is marked as a helper recovery beam candidate, denoted by $\mathcal{B}^{\mathrm{rec}}_{h,k}(t)$. A helper recovery beam candidate represents a helper beam whose coverage can include users affected by closed beams.

If a helper satellite h has at least one helper release beam candidate or one helper recovery beam candidate, the satellite is included in the helper satellite set $\mathcal{S}^{\mathrm{H}}(t)$.

3) Candidate Beam Relation Graph Construction: After the critical satellite set and helper satellite set are identified, the system constructs candidate beam relation graphs to record beam overlap relations. These graphs indicate which beam pairs have footprint overlap and can therefore form candidate reassociation relations. To make the graph construction easier to understand, this subsection uses the representative OneWeb-like local geometry shown in Fig. 2. In this illustrative example, each satellite is assumed to have 16 satellite-fixed downlink beams. Some beams of the critical satellites are shut down to satisfy the EPFD constraint, while neighboring satellites provide possible helper beams for service recovery. Based on the beam overlap relations in this local geometry, the system constructs the safe-release candidate graph and the closed-recovery candidate graph.

As shown in Fig. 3, the safe-release candidate graph is constructed from the overlap relations between critical safe beams and helper release beam candidates in the illustrative geometry. This graph records which critical safe beams have footprint

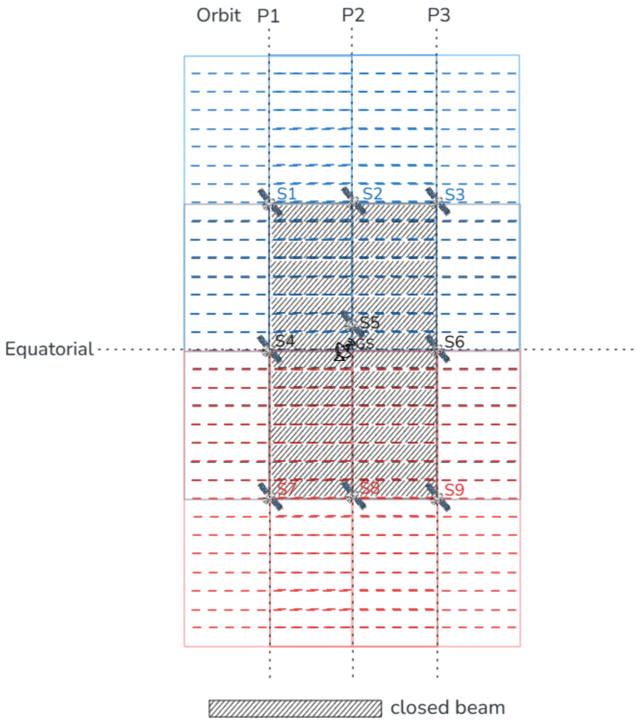


Fig. 2. Illustrative OneWeb-like local geometry used to explain candidate beam relation graph construction. Satellites S2, S5, and S8 are critical satellites, and S1, S3, S4, S6, S7, and S9 are helper satellites.

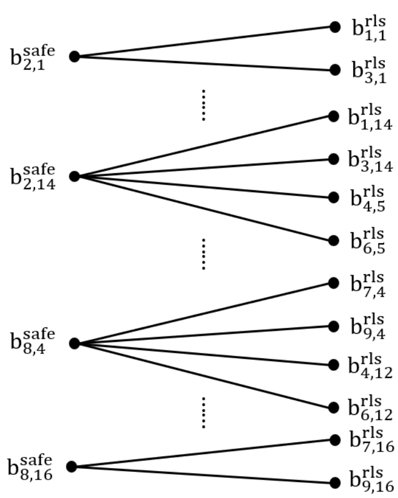


Fig. 3. Safe-release candidate graph between critical safe beams and helper release beam candidates. The left-side nodes $b^{\mathrm{safe}}_{s,\ell}$ denote safe beams on critical satellites, and the right-side nodes $b^{\mathrm{rls}}_{h,k}$ denote helper release beam candidates. An edge $e = (b^{\mathrm{safe}}_{s,\ell}, b^{\mathrm{rls}}_{h,k})$ exists when the two beams have footprint overlap.

overlap with helper release beam candidates. It provides the candidate beam pairs used in Safe-Beam Reassociation for Helper Users (SBR), and the detailed reassociation procedure is described in Section IV-C.

As shown in Fig. 4, the closed-recovery candidate graph is constructed from the overlap relations between critical closed beams and helper recovery beam candidates in the illustrative geometry. Each edge records a beam pair with

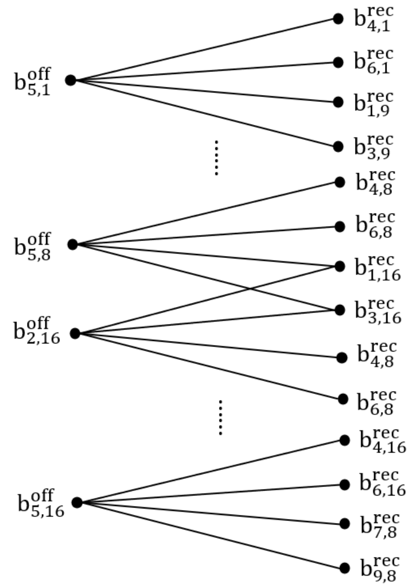


Fig. 4. Closed-recovery candidate graph between critical closed beams and helper recovery beam candidates. The left-side nodes $b^{\mathrm{off}}_{s,\ell}$ denote closed beams on critical satellites, and the right-side nodes $b^{\mathrm{rec}}_{h,k}$ denote helper recovery beam candidates. An edge $e = (b^{\mathrm{off}}_{s,\ell}, b^{\mathrm{rec}}_{h,k})$ exists when the two beams have footprint overlap.

footprint overlap. This graph provides the candidate beam pairs used in Helper-Beam Reassociation for Closed-Beam Users (HBR), and the detailed reassociation procedure is described in Section IV-E.

Through these two candidate beam relation graphs, the system records the beam pairs with footprint overlap between critical satellites and helper satellites. The safe-release candidate graph records the beam pairs between critical safe beams and helper release beam candidates, while the closed-recovery candidate graph records the beam pairs between critical closed beams and helper recovery beam candidates. These recorded beam pairs are used as candidates before users are reassociated.

It should be noted that the candidate beam relation graphs are used only to illustrate the footprint overlap relations between beams. They show which beam pairs may have overlap in the current time slot, but they do not directly determine the reassociation result. The actual reassociation decisions are made later in Safe-Beam Reassociation for Helper Users (SBR) and Helper-Beam Reassociation for Closed-Beam Users (HBR).

C. Safe-Beam Reassociation for Helper Users

Safe-Beam Reassociation for Helper Users (SBR) is the first reassociation procedure in the proposed framework. It is performed based on the safe-release candidate graph constructed in the previous subsection. The purpose of SBR is to let critical safe beams take over selected users originally served by helper release beams. After these users are reassociated to critical safe beams, the corresponding helper release beams can reduce their original service load. The released helper-side resources are then returned to the power budget of the helper satellite.

As shown in Fig. 3, the left-side nodes are critical safe beams, and the right-side nodes are helper release beam candidates. For example, the critical safe beam $b_{2,1}^{\mathrm{safe}}$ is connected to two helper release beam candidates, $b_{1,1}^{\mathrm{rls}}$ and $b_{3,1}^{\mathrm{rls}}$. This means that safe beam 1 of critical satellite S2 overlaps with release beam 1 of helper satellites S1 and S3. These beam pairs can be used as candidate reassociation choices in SBR.

1) Candidate Helper User Set Construction: For a helper release beam candidate, the users currently served by this beam are called helper-side users. For an edge e, not all helper-side users are relevant to SBR. The helper-side users that are also covered by the critical safe beam connected to the same edge can be considered for reassociation through this edge. These users form the candidate helper-user set of edge e, denoted by $\mathcal{U}_e^{\mathrm{cand}}(t)$. At this stage, $\mathcal{U}_e^{\mathrm{cand}}(t)$ only represents the users that may be considered for reassociation through edge e, and no user is reassociated yet.

2) Release Score Evaluation: For each edge e, SBR evaluates how much helper-side service demand can be moved from the helper release beam to the connected critical safe beam. The evaluation is performed over the candidate helper-user set $\mathcal{U}_e^{\mathrm{cand}}(t)$. From this set, SBR searches for a feasible user subset $\mathcal{U}_e^{\mathrm{sel}}(t)$ that can be taken over by the critical safe beam.

A candidate helper user can be included in $\mathcal{U}_e^{\mathrm{sel}}(t)$ only when two conditions are satisfied. First, after this user is moved from the helper release beam to the connected critical safe beam, its service satisfaction must not be lower than its original satisfaction on the helper release beam. If the user would receive lower satisfaction after reassociation, this user remains served by the original helper release beam and is not selected for this edge. Second, the critical safe beam must still have enough remaining capacity to serve the selected users. The total demand of the selected users in $\mathcal{U}_e^{\mathrm{sel}}(t)$ cannot be larger than the remaining capacity of the critical safe beam.

The release score of edge e is defined as the maximum total service demand that can be moved through this edge:

$$\Phi_e^{\mathrm{SBR}}(t) = \max_{\mathcal{U}_e^{\mathrm{sel}}(t) \subseteq \mathcal{U}_e^{\mathrm{cand}}(t)} \sum_{u \in \mathcal{U}_e^{\mathrm{sel}}(t)} D_u, \tag{4}$$

where $\mathcal{U}_e^{\mathrm{sel}}(t)$ is restricted to the feasible user subsets that satisfy the two conditions described above. A larger $\Phi_e^{\mathrm{SBR}}(t)$ means that more helper-side service demand can be moved away from the helper release beam through edge e. Therefore, this score is designed to prioritize the edge that can release the largest amount of helper-side service load while satisfying the capacity and non-degradation conditions. If no feasible user subset can be found, or if the connected critical safe beam has no remaining capacity, the release score of this edge is set to zero.

3) Iterative Edge Selection and Update: The safe-release candidate graph may contain several connected components. Each component includes the critical safe beams and helper release beam candidates that are connected through footprint overlap. SBR processes these components iteratively. In each component, SBR selects the edge with the highest release score. The selected edge is marked as an active safe-release relation, and the feasible user subset corresponding to this selected edge is reassociated from the helper release beam to the connected critical safe beam. After the reassociation, the loading and remaining capacity of the critical safe beam are updated, and the service load originally carried by the helper release beam is reduced. The released helper-side resources are added back to the power budget of the helper satellite.

As illustrated in Fig. 5, the selected edge is marked in green with a check mark. After one edge is selected, the release scores of the remaining edges in the same component are updated because the remaining capacity of the connected critical safe beam may have changed. If a critical safe beam no longer has enough remaining capacity to take over additional helper users, the remaining edges connected to this safe beam are assigned a release score of zero. SBR then selects the next edge with the highest release score in the component. This process continues until the remaining edges in each component have zero release scores. At the end of SBR, the graph contains the selected safe-release relations and the remaining zero-score edges. The corresponding helper release beams reduce their original service load, and the released helper-side resources are added back to the power budget of the helper satellite.

D. Released Power Allocation for Helper Recovery Beams

After SBR is completed, the transmit power released by helper release beams is added back to the available power budget of the corresponding helper satellite. The system then allocates this available power to helper recovery beams on the same helper satellite as the available power of helper recovery beams for serving closed-beam users.

To determine the power allocation order, the system first checks the candidate users connected to each helper recovery beam according to the closed-recovery candidate graph. These candidate users come from critical closed beams that have footprint overlap with the helper recovery beam. The system then calculates the priority-weighted recovery demand of each helper recovery beam based on the service priorities and traffic demands of these candidate users. If a helper recovery beam is connected to more high-priority users or users with larger traffic demand, the beam has a higher recovery demand value.

After the recovery demand value is calculated, the helper recovery beams are sorted in descending order of priority-weighted recovery demand. The released power is then allocated to the helper recovery beams according to this order. Each helper recovery beam can be increased at most to its beam power upper bound. If a beam reaches its upper bound, the remaining released power is allocated to the next helper recovery beam. This process continues until the released power is exhausted or all helper recovery beams have reached their beam power upper bounds.

E. Helper-Beam Reassociation for Closed-Beam Users

Helper-Beam Reassociation for Closed-Beam Users (HBR) is performed after helper recovery beams obtain available recovery resources. It is based on the closed-recovery candidate graph constructed in Section IV-B. The purpose of HBR is to reassociate users originally served by critical closed beams to neighboring helper recovery beams. Through this step, users affected by beam shutdown can receive service from helper satellites.

As shown in Fig. 4, the left-side nodes are critical closed beams, and the right-side nodes are helper recovery beam candidates. For example, the closed beam $b_{5,1}^{\mathrm{off}}$ is connected to several helper recovery beam candidates, such as $b_{4,1}^{\mathrm{rec}}$, $b_{6,1}^{\mathrm{rec}}$, $b_{1,9}^{\mathrm{rec}}$, and $b_{3,9}^{\mathrm{rec}}$. This means that users originally located under closed beam 1 of critical satellite S5 may also be covered by these helper recovery beams. These beam pairs can be used as candidate reassociation choices in HBR.

1) Candidate Closed-Beam User Set Construction: For a critical closed beam, the users originally served by this beam before shutdown are called closed-beam users. These users lose their original serving beam after the beam is shut down and therefore need to be reassociated to helper recovery beams. For an edge e in the closed-recovery candidate graph, not all closed-beam users can be served through this edge. The closed-beam users that are also covered by the helper recovery beam connected to the same edge can be considered for reassociation. These users form the candidate closed-beam user set of edge e, denoted by $\mathcal{U}_e^{\mathrm{rec,cand}}(t)$. At this stage, $\mathcal{U}_e^{\mathrm{rec,cand}}(t)$ only represents the users that may be recovered through edge e, and no user is reassociated yet.

2) Recovery Score Evaluation: For each edge e, HBR evaluates how much service can be recovered if some candidate closed-beam users are reassociated to the connected helper recovery beam. The evaluation is performed over the candidate closed-beam user set $\mathcal{U}_e^{\mathrm{rec,cand}}(t)$. From this set, HBR searches for a feasible user subset $\mathcal{U}_e^{\mathrm{rec,sel}}(t)$ that can be served by the helper recovery beam under its available recovery resources.

A candidate closed-beam user can be included in $\mathcal{U}_e^{\mathrm{rec,sel}}(t)$ only when the helper recovery beam still has enough available capacity to serve the selected users. Since the available recovery capacity of a helper recovery beam is limited, HBR first favors high-priority users and then selects the user subset that can achieve the largest recovered satisfaction under the available capacity. Therefore, the recovery score considers both the user priority and the estimated satisfaction after reassociation. The recovery score of edge e is defined as

$$\Phi_e^{\mathrm{HBR}}(t) = \max_{\mathcal{U}_e^{\mathrm{rec,sel}}(t) \subseteq \mathcal{U}_e^{\mathrm{rec,cand}}(t)} \sum_{u \in \mathcal{U}_e^{\mathrm{rec,sel}}(t)} w_{\pi(u)} \, \hat{s}_{u,e}^{\mathrm{rec}}(t), \tag{5}$$

where $\mathcal{U}_e^{\mathrm{rec,sel}}(t)$ is restricted to the feasible user subsets that can be supported by the connected helper recovery beam. Here, $w_{\pi(u)}$ is the priority weight of user u, and $\hat{s}_{u,e}^{\mathrm{rec}}(t)$ is the estimated satisfaction of user u if it is reassociated through edge e. A larger $\Phi_e^{\mathrm{HBR}}(t)$ means that this edge can recover a user subset with higher priority-weighted satisfaction. If no candidate closed-beam user can be served through this edge, or if the connected helper recovery beam has no available recovery capacity, the recovery score of this edge is set to zero.

3) Iterative Edge Selection and Update: The closed-recovery candidate graph may also contain several connected components. Each component includes the critical closed beams and helper recovery beam candidates that are connected through footprint overlap. HBR processes these components iteratively. In each component, HBR selects the edge with the highest recovery score. The selected edge is marked as an active closed-recovery relation, and the corresponding feasible user subset is reassociated from the critical closed beam to the connected helper recovery beam. After the reassociation, the remaining users under the critical closed beam are updated, and the remaining recovery capacity of the helper recovery beam is also updated.

As illustrated in Fig. 6, the selected edge is marked in green with a check mark. After one edge is selected, the recovery scores of the remaining edges in the same component are updated because the remaining users and the available recovery capacity may have changed. If a critical closed beam has no remaining user to recover, the remaining edges connected to this closed beam are assigned a recovery score of zero. Similarly, if a helper recovery beam has no remaining recovery capacity, the related edges are also assigned a recovery score of zero. HBR then selects the next edge with the highest recovery score in the component. This process continues until the remaining edges in each component have zero recovery scores. At the end of HBR, the framework obtains the selected closed-recovery relations and the closed-beam users that are moved to helper recovery beams. These users can then receive service from neighboring helper satellites.

F. Overall Procedure and State Update

This section summarizes the overall procedure of EPFD-Aware Beam Reassociation for Service Recovery. The previous subsections describe the time-slot beam role record, SBR, released power allocation, and HBR. This section connects these steps into a complete time-slot procedure and explains the system state that must be updated after each time slot.

Before the online reassociation procedures are executed, the system precomputes the time-slot beam role records for all time slots according to predictable satellite motion, beam footprints, user coverage, and beam-level EPFD contributions. If the aggregate EPFD of a time slot already satisfies the EPFD limit, beam shutdown is not required in that time slot. If the aggregate EPFD exceeds the limit, the corresponding critical satellites, closed beams, critical safe beams, helper release beam candidates, and helper recovery beam candidates are stored in the record.

The system further constructs the safe-release candidate graphs and the closed-recovery candidate graphs. The safe-release candidate graph is used as the input of SBR to determine which helper-side users may be reassociated to critical safe beams, thereby releasing helper satellite transmit power. The closed-recovery candidate graph is used as the input of HBR to determine which closed-beam users may be reassociated to helper recovery beams.

In each time slot, the system retrieves the corresponding time-slot beam role record. If no closed beam exists in the time slot, the system does not need to execute the following reassociation procedures. If closed beams exist, the system first performs SBR. SBR selects active safe-release relations according to the release score of each safe-release edge and reassociates the feasible helper-side user subsets to the corresponding critical safe beams. After SBR is completed, the loading of helper release beams is reduced, and the corresponding released transmit power is added to the available power budget of the helper satellite.

Before HBR is executed, the system performs released power allocation. This step allocates the available power of the helper satellite to helper recovery beams according to their priority-weighted recovery demand. After the allocation, each helper recovery beam obtains updated available recovery power.

Finally, the system performs HBR. HBR selects active closed-recovery relations according to the recovery score of each closed-recovery edge and reassociates the feasible closed-beam user subsets to the corresponding helper recovery beams.

Algorithm 1 Overall Procedure of EPFD-Aware Beam Reassociation for Service Recovery

Input: LEO constellation and beam configuration, GSO–GS and EPFD model, user distribution with service demand and priority weights, and beam power configuration
Output: User association sequence $\mathbf{X}$ and beam power allocation sequence $\mathbf{P}$

Initialize $\mathbf{X}(t)$ and $\mathbf{P}(t)$
Precompute the time-slot beam role records for all time slots
Construct the safe-release candidate graphs for all time slots
Construct the closed-recovery candidate graphs for all time slots
for each time slot t **do**
    Retrieve the time-slot beam role record at time slot t
    if no closed beam exists **then**
        continue
    end if
    // Safe-Beam Reassociation for Helper Users
    Initialize the active safe-release relation set as empty
    Evaluate $\Phi_e^{\mathrm{SBR}}(t)$ of each safe-release edge e by (4)
    while there exists a safe-release edge with $\Phi_e^{\mathrm{SBR}}(t) > 0$ **do**
        Divide the current safe-release candidate graph into connected components
        for each connected component **do**
            Select the edge e with the maximum $\Phi_e^{\mathrm{SBR}}(t)$
            Reassociate the feasible user subset in $\mathcal{U}_e^{\mathrm{cand}}(t)$ of edge e to the corresponding critical safe beam
            Add edge e to the active safe-release relation set
            Mark edge e as selected
        end for
        Update the safe-release candidate graph and recompute the release scores by (4)
    end while
    // Released Power Allocation for Helper Recovery Beams
    Compute the priority-weighted recovery demand of each helper recovery beam
    Sort helper recovery beams in descending order of priority-weighted recovery demand
    Allocate the released power to helper recovery beams subject to beam and satellite power limits
    Update the available power of helper recovery beams
    // Helper-Beam Reassociation for Closed-Beam Users
    Initialize the active closed-recovery relation set as empty
    Evaluate $\Phi_e^{\mathrm{HBR}}(t)$ of each closed-recovery edge e by (5)
    while there exists a closed-recovery edge with $\Phi_e^{\mathrm{HBR}}(t) > 0$ **do**
        Divide the current closed-recovery candidate graph into connected components
        for each connected component **do**
            Select the edge e with the maximum $\Phi_e^{\mathrm{HBR}}(t)$
            Reassociate the feasible user subset in $\mathcal{U}_e^{\mathrm{rec,cand}}(t)$ of edge e to the corresponding helper recovery beam
            Add edge e to the active closed-recovery relation set
            Mark edge e as selected
        end for
        Update the closed-recovery candidate graph and recompute the recovery scores by (5)
    end while
    Update $\mathbf{X}(t)$ and $\mathbf{P}(t)$
end for
return $\mathbf{X}$, $\mathbf{P}$

V. Discussion

A. Applicability to Different Multi-Beam LEO Constellations

The proposed EABR framework is designed for general multi-beam LEO satellite systems and is not limited to a specific constellation. Although the method and evaluation in this work adopts a OneWeb-like constellation as the simulation case, the core idea of EABR depends on predictable satellite motion, beam footprints, beam-level EPFD contributions, and coverage overlap among neighboring LEO satellites. These properties are common in many multi-beam LEO systems.

Therefore, as long as the satellite geometry and beam coverage can be estimated in advance, the time-slot beam role record and candidate beam relation graphs can be constructed for different constellation configurations. The proposed framework can then identify the critical satellites and helper satellites according to the current system state. In this sense, EABR is applicable to different multi-beam LEO constellations, while the OneWeb-like setting is used as a representative case for evaluation.

B. Helper Availability and Resource-Limited Recovery

The performance of helper-based service recovery depends on the availability of neighboring helper satellites and helper beams. In practical systems, helper availability is affected by constellation-level resource conditions, such as satellite density, orbital spacing, beam footprint overlap, EPFD limit strictness, and the remaining transmit power of neighboring satellites. Therefore, insufficient helper resources should be regarded as a system-level resource limitation rather than a failure of the reassociation framework.

When helper satellite resources are limited, the recovery performance of EABR is bounded by the available helper coverage, helper beam capacity, and helper satellite transmit power. Nevertheless, as long as helper coverage or helper power is available, EABR can still exploit these resources for service recovery and provide additional satisfaction improvement over conventional interference mitigation approaches. The framework also maintains the EPFD constraint and power constraints during the reassociation process, ensuring that service recovery does not introduce additional interference violation.

VI. Evaluation

A. Evaluation Overview

This chapter evaluates the performance of the proposed EPFD-Aware Beam Reassociation for Service Recovery (EABR) framework, focusing on its ability to improve user satisfaction and service continuity under the aggregate EPFD constraint. The evaluation consists of three main parts. First, we observe the aggregate EPFD variation during the critical period near the GS, as well as the number of beams that must be shut down to satisfy the EPFD constraint. This analysis confirms the existence of EPFD violation and beam shutdown demand during the critical period, and further supports the rationale of designing a reassociation-based service recovery

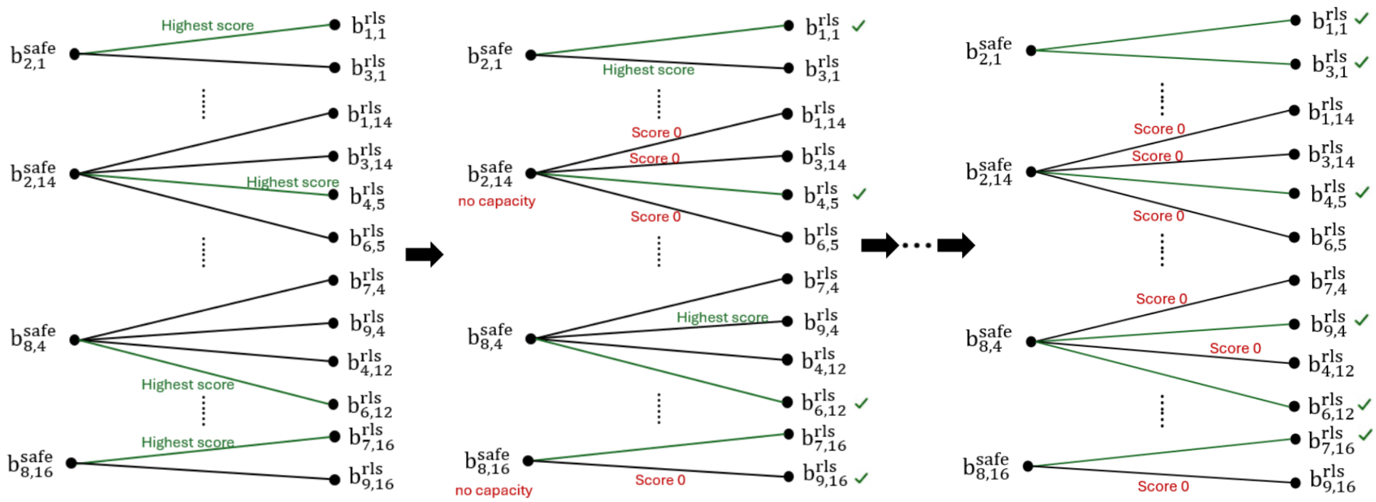


Fig. 5. Example of iterative edge activation and graph update in SBR.

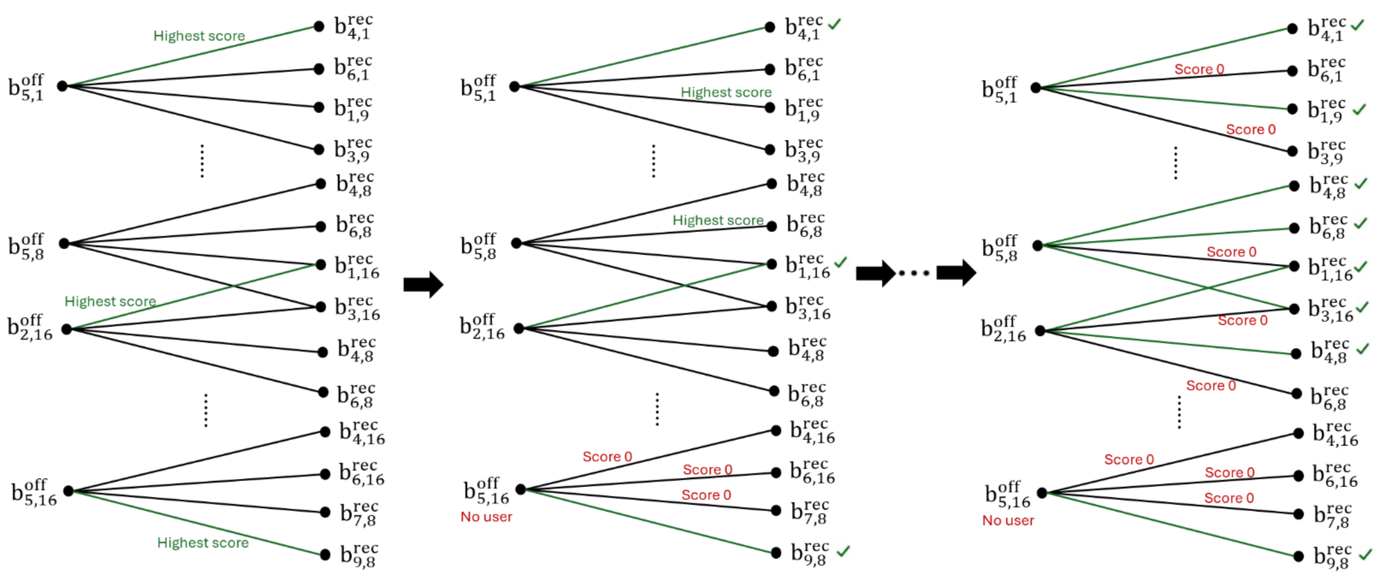


Fig. 6. Example of iterative edge activation and graph update in HBR. Green edges denote selected closed-recovery relations. After selected closed-beam users are reassociated to helper recovery beams, the available helper power and remaining unrecovered users are updated, and the recovery scores of the remaining candidate edges are recalculated.

framework. Second, we compare different interference mitigation methods in terms of service performance and service recovery capability. Finally, we further analyze the impact of different EPFD limits on user satisfaction to evaluate the stability and applicability of EABR under different interference constraint conditions.

B. Simulation Setup

To evaluate the performance of the proposed EABR framework under the NGSO–GSO coexistence scenario, we design a simulation framework that combines STK and MATLAB. This framework is used to construct the LEO constellation, obtain the time-varying satellite geometry, and further evaluate the performance of different interference mitigation methods in terms of EPFD compliance, downlink capacity, and user satisfaction.

The simulation uses the following two main tools:

- **STK (Satellite Tool Kit):** [25] STK is used to construct the OneWeb-like LEO constellation and generate the relative positions, visibility relationships, and beam coverage information among the LEO satellites, the GS, and users at each time slot.
- **MATLAB:** [26] MATLAB is used to process the system information generated by STK and perform offline preprocessing. This includes constructing the beam–user association, critical/helper satellite relationships, time-slot beam role records, and candidate beam relation graphs. MATLAB is then used to execute the proposed EABR framework and calculate the EPFD and user satisfaction performance of each method at different time slots.

The main simulation parameters used in this chapter are summarized in Table I. The table integrates the settings related

TABLE I
SIMULATION PARAMETERS.

Parameter	Value
LEO constellation	
Altitude (km)	1200
Orbit plane inclination	87.9°
Time slot	1 s
Maximum transmit power	16.8 W per satellite
Maximum antenna gain	34.33 dBi
Downlink frequency	11.7 GHz
Bandwidth	250 MHz
Antenna pattern	[19] $A \exp(\beta\phi)$
Maximum tilt angle	10°
Approximated antenna gain coefficient, (A)	4.0×10^3
Approximated antenna gain coefficient, (B)	-0.0671
GSO satellite	
Altitude (km)	35785.4
Longitude	30.6
Latitude	0
Downlink frequency	11.7 GHz
GSO ground station	
Longitude	30.6
Latitude	0
Minimum elevation angle	10°
Maximum receive gain	38.1 dBi
Noise temperature	240 K
Antenna pattern	[12]
LEO user ground station	
Maximum receive gain	40.95 dBi
Noise temperature	240 K
Antenna pattern	[12]
User demand	50 Mbps
EPFD	
EPFD limit (dB(W/m^2/1 MHz))	-173.4 [9]
Reference antenna diameter	60 cm

to the LEO constellation, GSO satellite, GS, LEO user ground station, and EPFD evaluation. These parameters are used in the following EPFD evaluation and user satisfaction comparison.

C. Compared Schemes

To evaluate the proposed EABR framework, we compare it with several interference mitigation methods. All methods are evaluated under the same simulation parameters to ensure a fair comparison. The compared schemes are described as follows:

- **Beam shutdown only:** When the aggregate EPFD exceeds the limit, the high-interference beams on critical satellites are directly shut down to satisfy the EPFD constraint. This method does not perform beam reassociation for service recovery.
- **PC + Tilt:** This method represents a power-control and satellite-tilting based interference mitigation strategy. It reduces the interference toward the GS by adjusting the transmit power and satellite tilt angle. Although this method can help satisfy the EPFD constraint, it does not provide a reassociation-based service recovery mechanism for users.
- **Only HBR:** This method only applies HBR. Helper recovery beams serve users located under critical closed beams using the default power of the helper satellite. SBR and released power allocation are not applied.
- **EABR (Proposed):** This method includes SBR, released power allocation for helper recovery beams, and HBR.

D. Evaluation Results

Based on the simulation setup and compared schemes described above, this section evaluates the performance of EABR during the critical period. The analysis is divided into four parts. First, the aggregate EPFD and the number of closed beams are examined to verify whether EABR can still satisfy the EPFD constraint after reassociation-based service recovery, and to observe the beam shutdown behavior of critical satellites at different time slots. Second, the average satisfaction of different methods is compared under different user loads to analyze how increasing system load affects service performance. Third, the number of recovered closed-beam users is used to explain the main reason why EABR improves satisfaction, and the satisfaction CDF is further used to observe the satisfaction distribution of all users. Finally, the average user satisfaction of EABR under different EPFD limits is evaluated to observe how the strictness of the interference limit affects service performance.

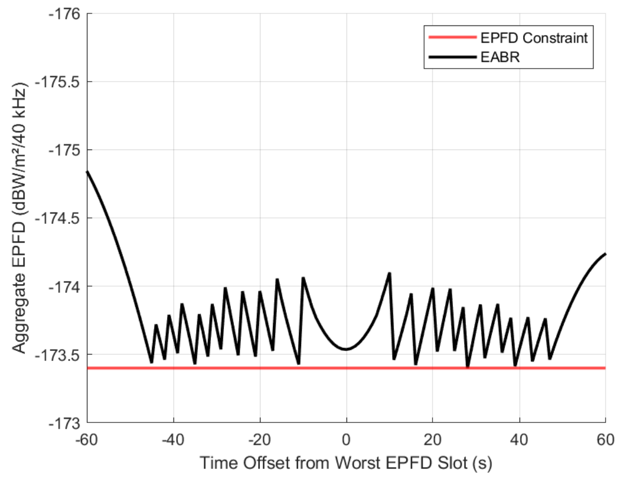


Fig. 7. Aggregate EPFD over time during the critical period.

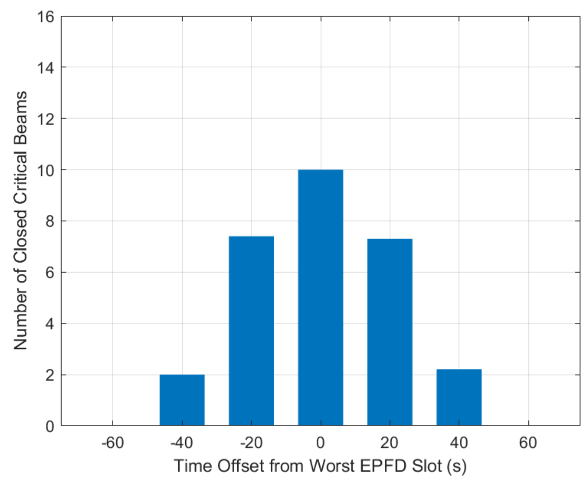


Fig. 8. Number of closed beams on critical satellites over time during the critical period.

1) EPFD Compliance and Beam Shutdown:

This subsection first analyzes the aggregate EPFD after applying EABR and further observes the number of beams that must be shut down on critical satellites to satisfy the

EPFD constraint. This analysis is used to verify whether the proposed framework can still satisfy the EPFD limit after reassociation-based service recovery, and to show the beam shutdown behavior during the critical period.

Fig. 7 shows the aggregate EPFD variation during the critical period near the GS. The red line represents the EPFD constraint, and the black line represents the aggregate EPFD after applying EABR. As shown in the figure, EABR keeps the aggregate EPFD below the limit during the entire observation period. This indicates that the proposed framework does not cause EPFD violation while performing service recovery for closed-beam users. Therefore, EABR can compensate for the service loss caused by beam shutdown while maintaining EPFD compliance.

Fig. 8 shows the number of beams that must be shut down on critical satellites at different time offsets. When critical satellites approach the worst EPFD slot, the number of closed beams gradually increases and reaches its maximum around the worst EPFD slot. After the satellites move away from the GS, the number of closed beams gradually decreases. This result shows that the beam shutdown demand changes with the satellite–GS geometry.

2) User Satisfaction under Different User Loads:

This subsection compares the average satisfaction of different interference mitigation methods under different user loads. By changing the number of users, we can observe how the service performance changes as the system load increases, and evaluate whether EABR can maintain user satisfaction under different load conditions.

Fig. 9 shows the average satisfaction under three user load scenarios: 30 users, 50 users, and 70 users. As shown in the figure, Beam shutdown only has the lowest satisfaction because it does not provide service recovery for users affected by closed beams. PC + Tilt can reduce the interference toward the GS, but it does not recover the service of users affected by beam shutdown. Only HBR can serve part of the closed-beam users through helper recovery beams, so it achieves better satisfaction than Beam shutdown only and PC + Tilt.

Compared with the other methods, EABR maintains higher average satisfaction under all three user loads. When the user load is low, the helper satellite usually has enough available power, so the difference between Only HBR and EABR is small. However, as the user load increases, more closed-beam users need service recovery, and Only HBR becomes limited by the default power of the helper satellite. In contrast, EABR uses SBR to release available power from the helper satellite and improve the service recovery capability of helper recovery beams. Therefore, EABR can maintain more stable service performance under medium and high user loads.

3) Service Recovery Capability and Satisfaction Distribution:

This subsection further analyzes why EABR improves user satisfaction and observes the satisfaction distribution of different methods. Since 30 users is a low-load scenario, the default available power of the helper satellite is usually sufficient to support most recovery demand. Therefore, the difference in service recovery capability is not obvious under 30 users. To better show the effect of the proposed framework, this subsection focuses on the 50-user and 70-user scenarios.

Fig. 10 compares the number of successfully recovered closed-beam users between Only HBR and EABR under the 50-user and 70-user scenarios. As shown in the figure, EABR serves more closed-beam users in both load scenarios. The difference becomes more obvious around the worst EPFD slot, where the beam shutdown demand is the most severe. This result shows the necessity of SBR in EABR.

Fig. 11 shows the satisfaction CDF of three interference mitigation methods. This figure focuses on the user-level satisfaction distribution. A CDF curve that shifts further to the right means that more users maintain higher satisfaction. In contrast, if the curve rises quickly in the low-satisfaction region, more users suffer from service degradation.

As shown in the figure, the CDF curve of EABR shifts more toward the high-satisfaction region than the other methods. This means that more users can maintain better service quality. The result shows that EABR not only improves average satisfaction, but also reduces the proportion of low-satisfaction users, making the overall user service quality more stable.

4) Impact of Different EPFD Limits:

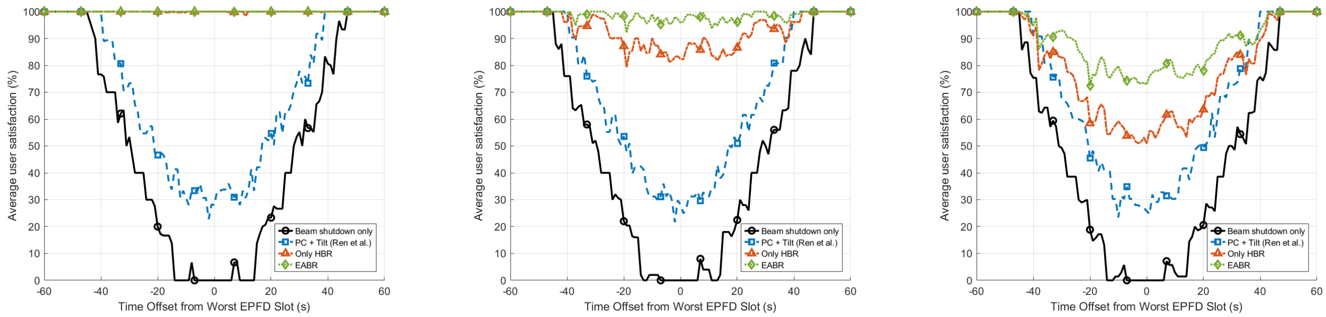
This subsection further evaluates the impact of different EPFD limits on the service performance of EABR. When the EPFD limit becomes stricter, the allowed aggregate interference becomes lower. As a result, critical satellites may need to shut down more beams to satisfy the constraint, and the recovery demand of closed-beam users may also increase. This analysis is used to observe whether EABR can maintain stable average user satisfaction under different EPFD limits.

Fig. 12 shows the average user satisfaction of EABR under different EPFD limits. As shown in the figure, when the EPFD limit is more relaxed, EABR can maintain higher average user satisfaction. This result shows that the strictness of the EPFD limit directly affects user satisfaction. It also indicates that EABR can achieve better service performance under more relaxed interference constraints.

Overall, EABR keeps the aggregate EPFD below the limit during the critical period and uses SBR to improve the ability of helper recovery beams to serve closed-beam users. Compared with other interference mitigation methods, EABR maintains higher average satisfaction under different user loads and reduces the proportion of low-satisfaction users. In addition, the results under different EPFD limits show that the strictness of the interference constraint affects user satisfaction, and EABR can maintain better service performance under more relaxed constraints. Overall, EABR achieves a better balance between EPFD compliance and service continuity.

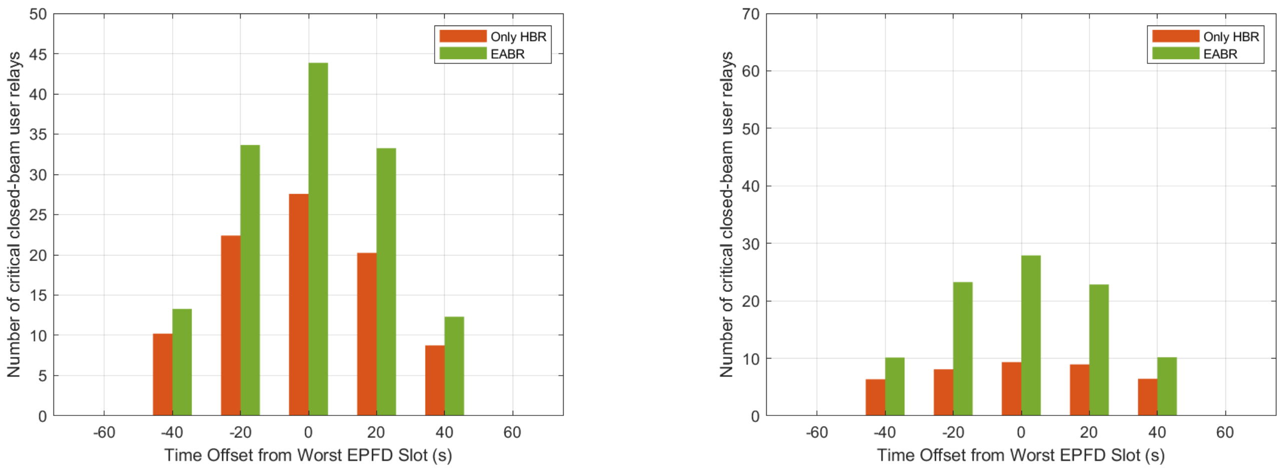
VII. Conclusion

This thesis proposed EPFD-Aware Beam Reassociation for Service Recovery (EABR), a beam reassociation framework for EPFD-constrained multi-beam LEO satellite systems under NGSO–GSO coexistence. EABR exploits predictable satellite motion, beam footprints, and coverage overlap among neighboring LEO satellites to support users affected by critical-beam shutdown. The framework consists of Safe-Beam Reassociation for Helper Users (SBR), released power allocation for



(a) (b) (c)

Fig. 9. Average user satisfaction of different interference mitigation methods under different user loads. (a) 30 users. (b) 50 users. (c) 70 users.



(a) (b)

Fig. 10. Number of recovered closed-beam users with and without SBR under different user loads. (a) 50 users. (b) 70 users.

helper recovery beams, and Helper-Beam Reassociation for Closed-Beam Users (HBR).

The simulation results show that EABR keeps the aggregate EPFD below the required limit during the critical period. Compared with beam shutdown only, PC + Tilt, and Only HBR, EABR achieves higher average user satisfaction under different user loads. The results also show that SBR increases the number of recovered closed-beam users, especially around the worst EPFD slot, and the satisfaction CDF confirms that EABR reduces the proportion of low-satisfaction users.

Overall, EABR provides a practical service recovery approach for multi-beam LEO satellite systems under the GS EPFD constraint. By using precomputed beam relation information and neighboring helper satellites, the proposed framework improves service continuity with a practical time-slot-based decision process. Therefore, EABR achieves a better balance between EPFD compliance, user satisfaction, and implementation complexity.

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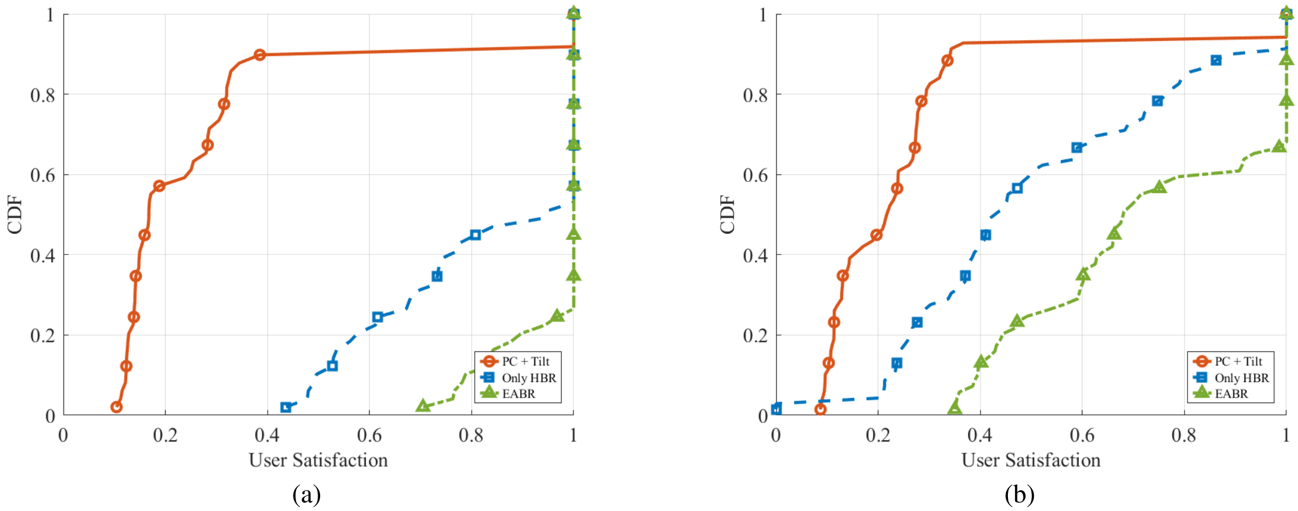

Fig. 11. CDF of user satisfaction for different interference mitigation methods under different user loads. (a) 50 users. (b) 70 users.

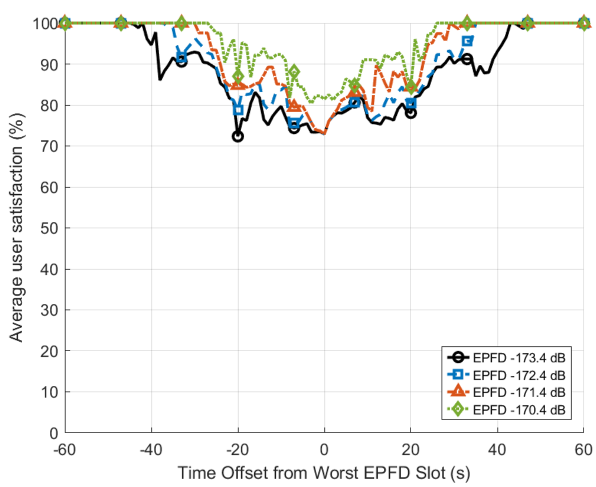

Fig. 12. Average user satisfaction of EABR under different EPFD limits.

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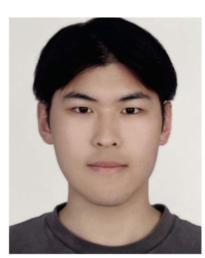

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